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Editors

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Dear Editors,

Please consider our manuscript, “Surrogate-assisted local conditional distribution inference with certified simultaneous gains”, by Jian Hou, Tan Meng and Maozai Tian, as a regular research article.

The paper certifies gains in simultaneous distributional inference from scarce reference measurements and broadly observed surrogates. A valid distributional example shows that minimising integrated variance can increase a simultaneous critical value. Pairing the candidate and Anchor critical values yields covariance-sensitivity and oracle bounds proportional to the amount borrowed, allowing nonunique optima. We distinguish finite-sample covariance-set certification from implementable asymptotic multiplier calibration, with separate protection for Monte Carlo quantile error. A local binary experiment identifies the information required to certify weak improvements.

The numerical study comprises 6,000 primary replications and 6,300 additional design-grid replications, including a matched paired-unpaired ablation, 101-threshold validation, non-Gaussian and discrete outcomes, two-dimensional localisation and nuisance-regression sensitivity. In the air-monitoring application, historical collocations train a frozen mean calibration, and subsequent observations define a distributional audit population. Independent designed verification samples assess humidity-specific exceedance discrepancies without treating temporal observations as independent population draws. The historical reference cost is reported separately from the subsequent audit budget. A further 1,600 budget-mask evaluations quantify upper-tail precision through a 95-percent verification budget without changing the historical fits or inferential target.

The manuscript and Supplement provide the method, proofs and numerical results. The accompanying frozen reproduction archive contains data, source-verified dates, code, replication-level results and build commands. The manuscript identifies the public repository for the original materials.

An earlier version was declined by JASA on novelty grounds. The present revision adds the simultaneous-gain certification problem, paired perturbation and information results, a guarded numerical implementation and a new dated application analysis. It is submitted here as a new manuscript.

The authors declare no conflicts of interest. ChatGPT assisted methodological development, mathematical drafting, programming, numerical checks and editorial preparation; earlier development also used OpenAI Codex. This assistance is disclosed in the manuscript, and responsibility for the work remains with the authors.

Yours sincerely,

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